

TASK MANAGEMENT SYSTEM

Full-Stack Technical Architecture, System Design & Interview Guide

1. Executive Summary & Technology Stack

The Task Management System is an enterprise-grade full-stack web application designed for organization-wide task delegation, workload monitoring, real-time status tracking, and transactional email alerts. Built using JavaScript (ES Modules) in an npm workspace monorepo architecture.

- **Monorepo Framework:** npm Workspaces managing decoupled server (Express API) and web (React Vite) applications.
- **Frontend:** React (Vite) + Tailwind CSS + TanStack Query (v5) + Lucide Icons + Plus Jakarta Sans font.
- **Backend Framework:** Express.js (Node.js) with Zod validation middleware and async handler wrappers.
- **Database & ORM:** Drizzle ORM paired with Neon Serverless PostgreSQL with pooled connections and SSL security.
- **Authentication & RBAC:** Better Auth adapter, Bcrypt password hashing, session tokens, and Role-Based Access Control (Admin/Employee).
- **Email Notifications:** Nodemailer transactional email dispatcher with non-blocking async execution.

2. Database Schema & Data Modeling

The PostgreSQL database contains 5 core tables and 3 custom enums managed strictly via Drizzle ORM.

PostgreSQL Custom Enums

- **role:** 'admin' | 'employee'
- **priority:** 'high' | 'medium' | 'low'
- **status:** 'not_started' | 'in_progress' | 'completed'

Core Table Structures

- **users Table:** id (UUID), name (Text), email (Unique Text), password_hash (Text), role (role Enum), department (Text), is_active (Boolean).
- **tasks Table:** id (UUID), title (Text), description (Text), priority (Enum), status (Enum), assigned_to_id (FK -> users.id), assigned_by_id (FK -> users.id), due_date (Timestamp).
- **sessions Table:** id (Text), user_id (FK -> users.id), token (Unique Text), expires_at (Timestamp).

3. Authentication, Security & Workflows

- **Bcrypt Password Hashing:** User passwords are salted and hashed prior to database persistence.
- **Session Management:** Cryptographically generated 64-character hex tokens stored in sessions table and validated per HTTP request.
- **Security Error Contracts:** Login failures return HTTP 401 'Invalid email or password' without revealing account existence.
- **Role-Based Middleware:** requireAuth verifies valid token/cookie, requireRole(['admin']) restricts executive actions.

4. REST API Endpoint Specification

- **POST /api/auth/login:** Public. Authenticates user & returns token + user profile.
- **GET /api/auth/me:** Authenticated. Validates active session token.
- **GET /api/employees:** Admin Only. Retrieves directory of all employees.
- **POST /api/employees:** Admin Only. Creates employee account & triggers welcome email.
- **GET /api/tasks/stats:** Admin Only. Computes real-time status counter metrics.
- **GET /api/tasks:** Authenticated. Lists tasks with SQL search, status/priority filters, and pagination.

- **POST /api/tasks:** Admin Only. Assigns task to employee & triggers notification email.
- **PATCH /api/tasks/:id/status:** Authenticated. Updates task status (optimistic UI on frontend).

5. Interview Technical Preparation & Q&A Cheatsheet

Q1: Why choose an npm workspace monorepo?

Answer: Keeps frontend and backend in a unified repository for developer productivity while permitting independent production builds and deployments (Express on Render, React Vite on Vercel).

Q2: Why TanStack Query over basic useEffect fetching?

Answer: Handles server-state caching, deduplication, loading/error states out of the box, and provides optimistic UI updates for instant task status changes without network lag.

Q3: How does backend pagination and search work?

Answer: Implemented at the database level using Drizzle ORM SQL queries (ilike pattern matching, limit, and offset) executed alongside parallel Promise.all total count queries.

Q4: How do email notifications stay performant?

Answer: Email dispatch is non-blocking. Database transactions complete first and respond to the client immediately, while emails trigger asynchronously in the background.